INFOTACT SOLUTIONS · DATA ANALYTICS

CareFlow

Clinical Pathway Process Mining

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THE PROBLEM

A hospital sees one average wait, not the reason for it

Without process mining

One average waiting time. The order of steps stays hidden, so loop-backs and stuck points stay hidden too.

With CareFlow

The real path of every patient, rebuilt from timestamps: which step follows which, and where time is lost.

From simulated records to a dashboard

1

Python generator

Simulates 10,000 patients and 68,440 events, then runs 9 quality checks

2

BigQuery

Stores the raw event log: 68,440 rows, 10,000 patients

3

dbt

Cleans it into an event log with 3 models and 11 tests

4

PM4Py

Discovers the real patient paths as a directed graph

5

Power BI

Shows bottlenecks and key numbers to a manager

PM4Py reads either the CSV or the dbt view in BigQuery. Both draw the identical process map, so the data was not changed along the way.

Three problems planted on purpose, so the tools have something to find

40%

Loop-back

of X-Ray patients are sent back to Triage before the doctor review

+20-60

Busy-hour doctor wait

extra minutes at 10:00 to 14:00 and 18:00 to 22:00, against 0 to 10 at other hours

+15-45

Slow evening pharmacy

extra minutes at 18:00 to 23:00, against 0 to 5 at other times

The rules are in the design document and coded in the generator. A fixed seed (42) makes every run give the same data.

The data was checked before anything was built on it

9/9

Validation checks pass

Python script on the CSV: counts, order of steps, no missing values

11

dbt tests pass

Run on the BigQuery views every time they are rebuilt

965

Same answer, two tools

dbt and PM4Py both count 965 loop-back patients

Measured loop-back rate: 965 of 2,377 X-Ray patients, 40.6%, close to the 40% planted.

PROOF · DATA QUALITY

validate_data.py: 9 of 9 checks pass

Run on data/event_log.csv: 68,440 events, 10,000 patients. Every rule from the design document is tested.

```
PS C:\...\careflow-clinical-pathway-process-mining> python
data_generator/validate_data.py

Checking 68440 events for 10000 patients

PASS  No missing values
PASS  Starts with Registration, ends with Discharge
PASS  Timestamps never go backwards
PASS  Ward only on Ward Admission rows
PASS  Every Ward Admission row has a Ward
PASS  Admitted patients skip Pharmacy and Billing
PASS  Activity names are valid
PASS  Condition and Age_Group constant per patient
PASS  No duplicate rows

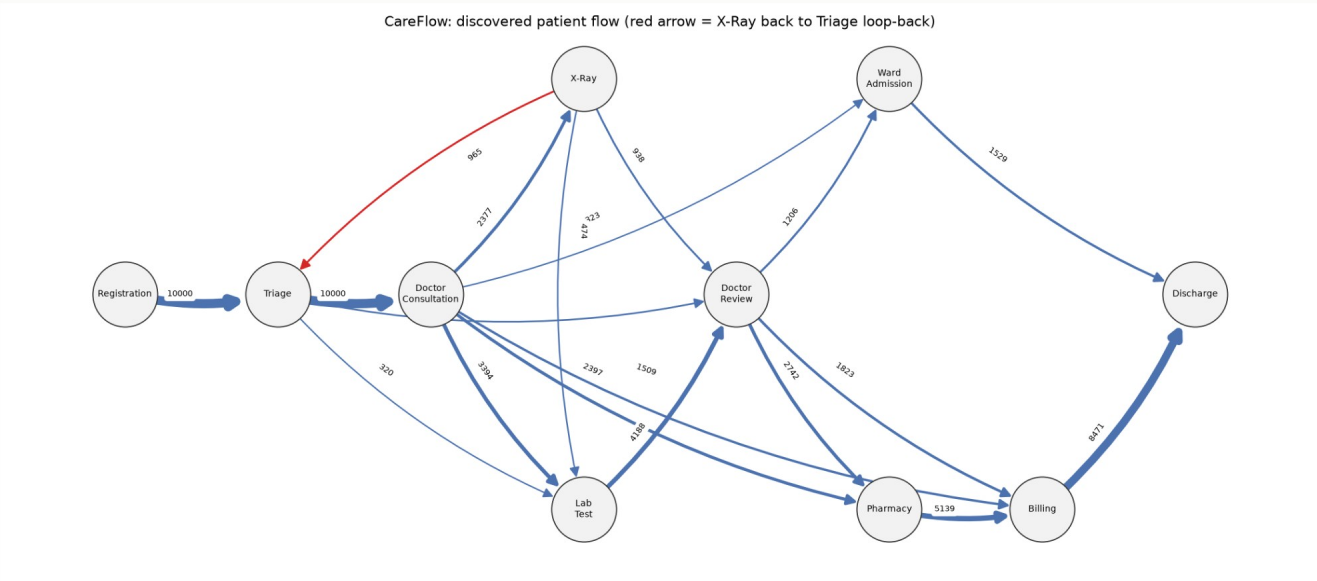
9 of 9 checks passed
```

PROCESS MAP

The loop-back is visible in the map

PM4Py found 18 different patient paths and 19 connections between steps.

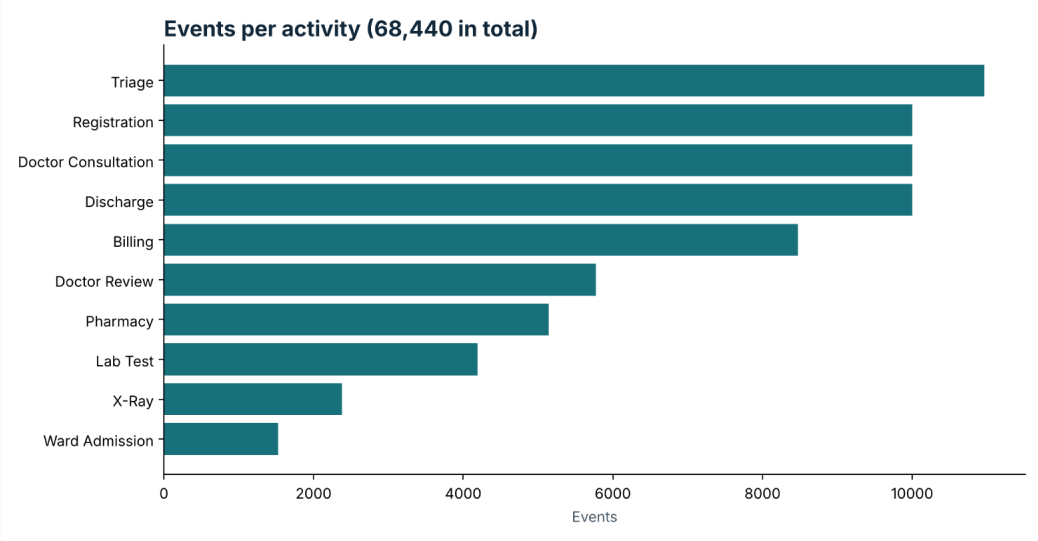
The arrow from X-Ray back to Triage is the planted loop-back.



PROOF · PATIENT PATHS

18 paths exist, but five cover most patients

Path after Doctor Consultation	Patients
Pharmacy, Billing	2,400
Lab Test, Review, Pharmacy, Billing	1,660
Billing only	1,510
Lab Test, Review, Billing	1,100
Lab Test, Review, Ward Admission	640

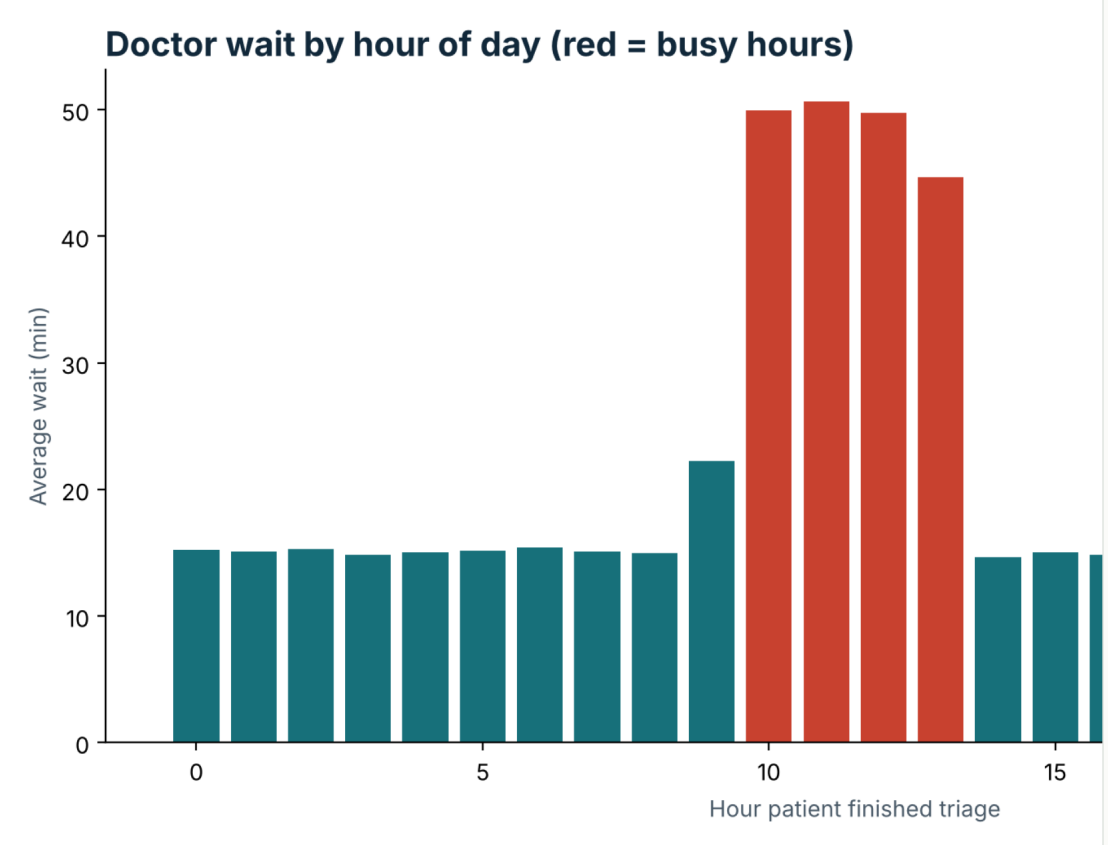


Where patients wait longest between steps

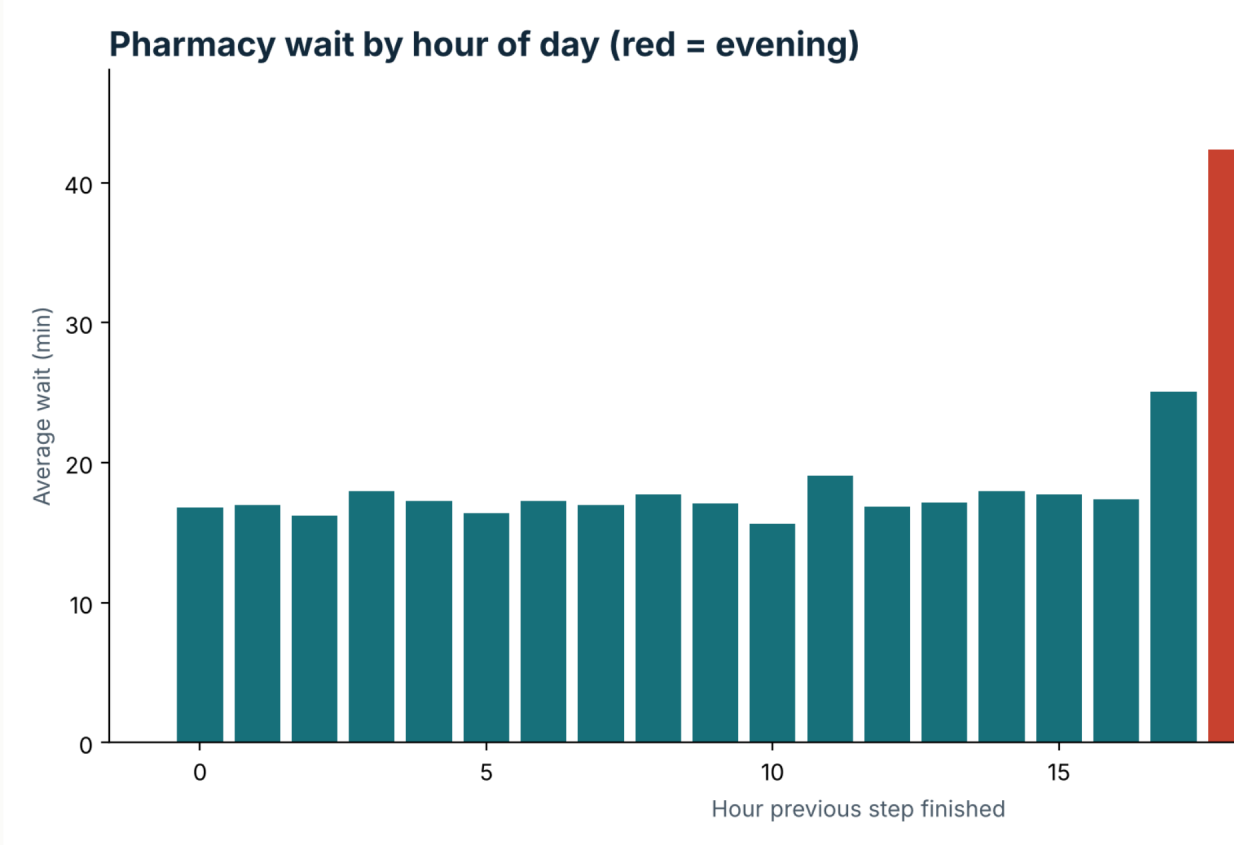
Transition	Patients	Avg wait (min)
Lab Test to Doctor Review	4,188	**40.0**
Doctor Consultation to Pharmacy	2,397	**28.4**
Triage to Doctor Consultation	10,000	**26.6**
X-Ray to Triage (loop-back)	965	**20.3**

Doctor waits average 48.5 min in busy hours against 17.8 min otherwise. Evening pharmacy averages 44.5 min against 18.2 min. Ward Admission to Discharge (298 min) is long by design, so it is left out.

Proof: waits jump exactly in the planted hours



Doctor: 48.5 min busy vs 17.8 min other

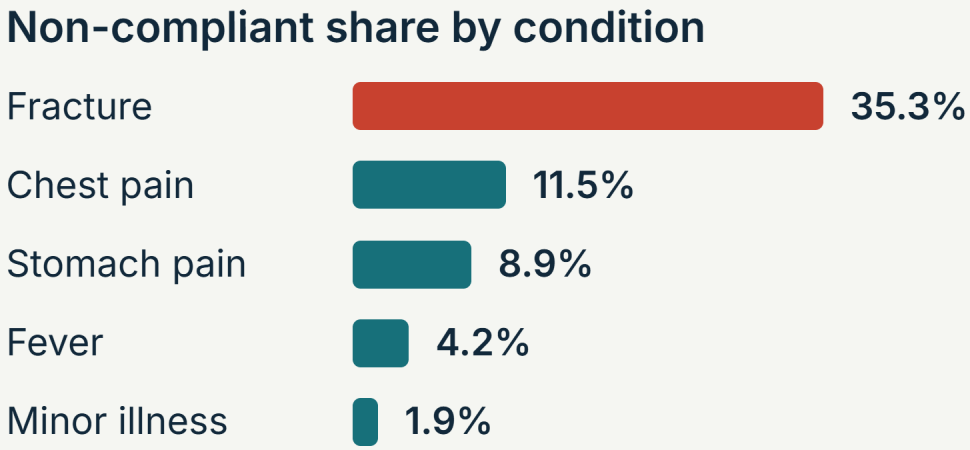


Pharmacy: 44.5 min evening vs 18.2 min other

Nine in ten patients follow the ideal path

90.35%

9,035 of 10,000 patients are compliant. The 965 who are not are exactly the loop-back patients.

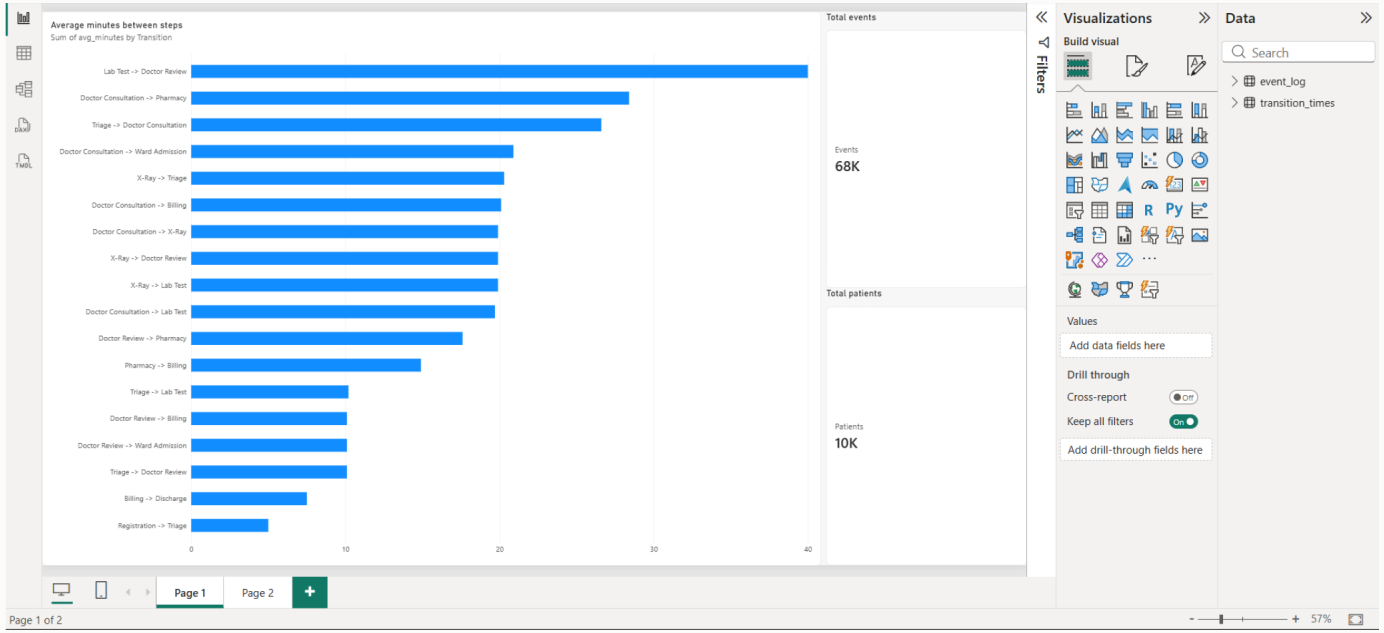


Conformance compares each patient's steps with the ideal order. Fracture patients need X-Rays most, so they loop back most.

POWER BI DASHBOARD

A manager sees the bottlenecks at a glance

Built from the dbt views in BigQuery, with key numbers and waiting times per step.



PROOF · GITHUB

All work is in a public repository

Commits on 22 different days. The repo holds the generator, BigQuery schema, dbt models, PM4Py script, Power BI file and docs.

github.com/charansai-popuri/careflow-clinical-pathway-process-mining

```
PS C:\...\careflow-clinical-pathway-process-mining> git log --oneline -12
e155af1 Record conformance results in README and update design progress
e76a69d Add dbt conformance model flagging paths that break the ideal journey
8c56fc2 Fix weekly progress formatting and add week 3 and bigquery run step
803d3fe Read the dbt event log from BigQuery in PM4Py and add week 3 progress
8efebae Add Power BI dashboard with bottleneck chart and process map
a58931d Document how to run dbt and record the dbt setup
c723487 Update docs, requirements and gitignore for the dbt models
c802394 Add dbt transition times model for bottleneck analysis
95c0031 Ignore dbt generated files and remove them from the repository
c5e81dd Run dbt model and tests successfully on BigQuery
aa1e98b Add dbt project with event log cleaning model and tests
4577583 Add weekly progress record for weeks 1 to 3
```

What a hospital would do with this

Fix the loop-back

Check why 40% of X-Ray patients return to Triage, starting with fracture cases.

Staff for busy hours

Add doctors at 10:00 to 14:00 and 18:00 to 22:00, when waits roughly triple.

Support the evening pharmacy

Extra pharmacy staff after 18:00 would cut the 44.5 min average wait.

Limits: the data is simulated, so these show the method works. Next, run the same pipeline on real hospital records.